Frontend Hosting Architecture using AWS

Powered by VPC, EC2, Route 53, ACM, ELB, ASG, and CloudFront

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Overview of the Architecture

VPC

Private, controlled network environment

EC2 + ELB

App hosted with load balancing

ACM

Provides SSL for secure HTTPS access

Auto Scaling

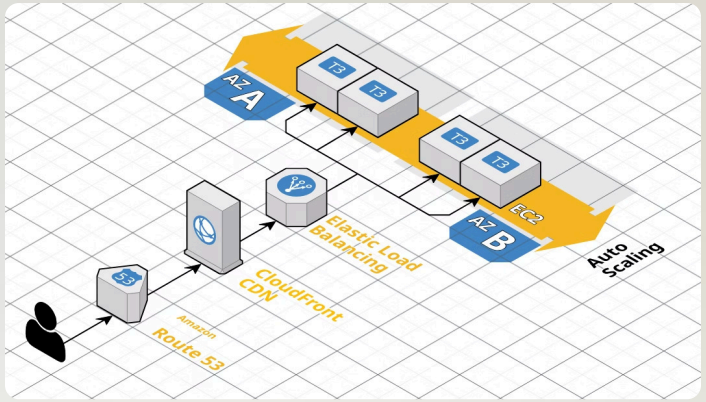
Adjusts capacity based on traffic

Route 53

Domain routing and DNS management

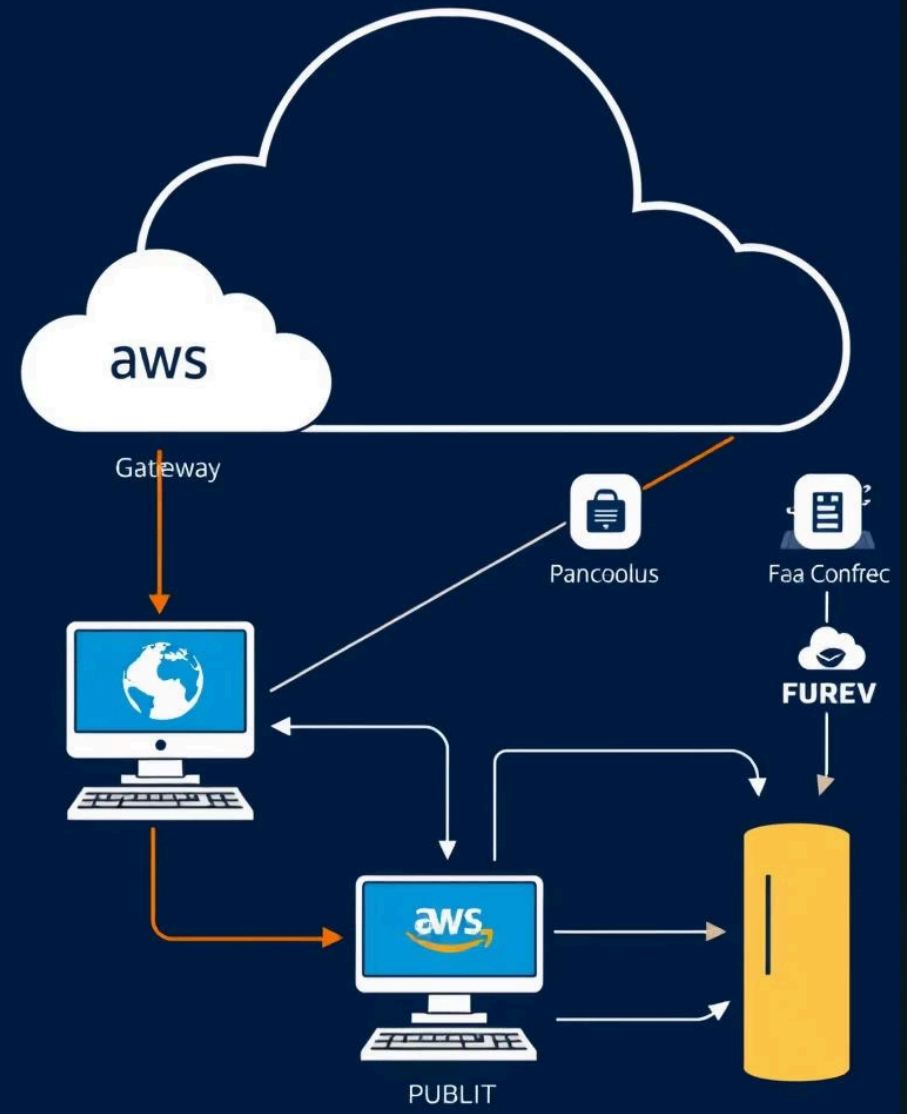
CloudFront

Global content delivery network



Architecture Diagram

Visual overview integrating all core AWS components



VPC and Networking

Custom VPC

Isolated and secure virtual network

Public Subnet

Hosts load balancers and gateways

Private Subnet

Hosts EC2 instances securely

Internet Gateway

Provides controlled internet access

EC2 and Auto Scaling Group

EC2 Hosts

Deploy frontend application code

Auto Scaling Group

Ensures availability and scalability

Custom AMI

Standardizes instance configuration

Launch Template & Health Checks

Automates instance management and monitoring

Elastic Load Balancer

Load Balancer Role

Efficiently routes incoming traffic

ACM Integration

Provides secure HTTPS connections

Target Group

Directs traffic to healthy EC2 instances

CloudFront (CDN)

CloudFront CDN

Caches static content worldwide

Positioned Before ALB

Serves cached content instantly

Speeds Up Delivery

Reduces latency for global users

Benefits of the Architecture

High Availability

Redundant resources and auto healing

Decoupled Frontend

Scalable and modular architecture

Auto Scaling

Handles variable traffic loads

Secure HTTPS

Encryption with ACM certificates

Global Content Delivery

CloudFront caches globally

Conclusion & Key Takeaways

Scalability

Auto Scaling adapts capacity in real time

Security

ACM SSL and VPC isolation protect data

High Availability

Load Balancer and Route 53 ensure uptime

Global Performance

CloudFront reduces latency worldwide

Cost Efficiency

Dynamic resource management reduces spend